

Universal Core Spacing in Molecular Cloud Filaments: Complete HGBS Analysis with Physical Mechanism Constraints

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Abstract

We present the most comprehensive analysis of core spacing along filaments in the Herschel Gould Belt Survey (HGBS) to date, including all 9 regions (5,411 cores) and the previously unstudied W3 region. Through standardized DisPerSE skeleton processing, we measure a universal core spacing of 0.213 ± 0.007 pc (statistical), corresponding to $2.13\times$ the characteristic filament width of 0.10 pc. When accounting for projection effects, the true 3D spacing is ~ 0.27 pc, or $\sim 2.7\times$ the filament width.

We address the fundamental question: why do filaments fragment at $\sim 2\text{--}3\times$ their width rather than the classical $4\times$ prediction? We combine observational analysis with linear perturbation theory, performing 3,240 simulations across the physical parameter space of finite-length effects, external pressure, magnetic fields, non-cylindrical geometry, and mass accretion. We demonstrate that **plausible combinations** of these physical effects—consistent with observed HGBS parameters—can explain the observed spacing range of $2\text{--}3\times$ the filament width.

To test the role of magnetic fields, we perform isothermal MHD turbulence simulations using the Athena++ code at Mach numbers 1 and 3 with plasma $\beta = 1$. The saturated states yield Alfvén speeds $v_A \approx 1.5\text{--}1.8 c_s$, suggesting that isotropic magnetic pressure support would *increase* the predicted fragmentation scale, moving it *away* from observations. This rules out isotropic magnetic pressure enhancement as the explanation and favors geometric effects—finite filament length, external pressure confinement, and mass accretion—as the primary mechanisms setting the $\sim 2\text{--}3\times$ core spacing scale.

1 Introduction

1.1 The Filament Paradigm

The *Herschel* Space Observatory has established that filaments are the fundamental structures within which star formation occurs in molecular clouds [André et al., 2010]. Two key observational results have emerged:

- **Characteristic filament width:** ~ 0.1 pc across diverse environments [Arzoumanian et al., 2011]
- **Critical mass-per-unit-length:** $M_{\text{line,crit}} \approx 16 M_{\odot}/\text{pc}$ [Ostriker, 1964]

Classical fragmentation theory [Inutsuka and Miyama, 1992] predicts that isothermal gas cylinders fragment at a wavelength of approximately $4\times$ the filament width. For the characteristic HGBS filament width of 0.10 pc, this predicts a core spacing of ~ 0.4 pc.

However, HGBS analyses have consistently measured core spacings of ~ 0.2 pc. Previous studies focused on 4 regions with robust sample sizes. In this work, we extend the analysis to **all 9 HGBS regions** including the previously unstudied W3 region, providing the most complete HGBS analysis to date. The sub-Jeans spacing phenomenon is not unique to HGBS; independent studies of the California molecular cloud also report core spacings of ~ 0.15 pc ($\sim 1.25\times$ filament width) significantly shorter than the classical $4\times$ prediction [Zhang et al., 2024]. Recent high-resolution studies have revealed hierarchical fragmentation from filaments to fibers to condensations, with fiber-level spacings consistent with classical theory [Yang et al., 2024].

1.2 The $2\times$ vs. $4\times$ Question

The observed $\sim 2\text{--}3\times$ spacing differs from the classical $4\times$ prediction by a factor of nearly 2. Several explanations have been proposed:

- **Finite length effects:** Real filaments are not infinite cylinders [Inutsuka and Miyama, 1997]
- **External pressure:** Gould Belt clouds are pressure-confined [Fischera and Martin, 2012]
- **Non-cylindrical geometry:** Real filaments show tapering and non-cylindrical shapes
- **Mass accretion:** Filaments grow by accreting material over time [Heitsch, 2013, Chen et al., 2024]

- **Turbulent compression:** Supersonic turbulence can compress scales [Padoan and Nordlund, 2011]
- **Sonic scale effects:** The sonic scale may set characteristic fragmentation length [Federrath, 2016]

We investigate these effects through comprehensive observational analysis and linear perturbation theory, supplemented by qualitative MHD turbulence simulations to assess magnetic field effects.

2 Observational Data and Methods

2.1 Complete HGBS Sample

Our sample includes all 9 HGBS regions, spanning a factor of 3 in distance and orders of magnitude in star formation activity:

Table 1: Complete HGBS Sample with All 9 Regions

Region	Distance (pc)	Total Cores	Prestellar Fraction (%)	Environmental Classification
Orion B	260	1,844	42.3	Active, High
Aquila	260	749	62.6	Very Active, High
Perseus	260	816	49.4	Active, High
Ophiuchus	130	513	28.1	Moderate, Medium
Serpens	260	194	26.8	Low, Low
TMC1	140	178	24.7	Low-Moderate, Low
CRA	130	239	9.6	Quiescent, Low
Taurus	140	536	9.7	Quiescent, Medium
W3	400	342	35.2	Active, Very High
Total		5,411		

W3 region: Located at 400 pc distance, W3 is a high-mass star-forming region with active massive star formation. Its inclusion provides a crucial high-pressure, high-extinction test of theoretical predictions. This is the first reported core spacing measurement for W3. A detailed discussion of W3 data processing and quality assessment is provided in §3.5.

2.2 DisPerSE Pipeline and Parameter Selection

Core extraction was performed using the DisPerSE (Discrete Persistent Structures Extractor) algorithm [Sousbie, 2011], which identifies filamentary structures based on persistence topology in the column

density maps. For our HGBS analysis, we adopted standardized pipeline parameters consistent with published HGBS analyses:

- **Persistence threshold:** 3σ above local noise, where σ is the standard deviation of the background column density fluctuations
- **Robustness threshold:** 5 (dimensionless persistence parameter)
- **Smoothing scale:** Gaussian filtering with FWHM equal to the Herschel SPIRE $250\ \mu\text{m}$ beam ($18''$)
- **Column density threshold:** $A_V \geq 2$ mag (equivalent to $N_{\text{H}_2} \approx 2 \times 10^{21}\ \text{cm}^{-2}$)

Distance scaling: For regions spanning distances from 130–400 pc, the same angular resolution thresholds correspond to different physical scales. To maintain consistency, we applied a distance-dependent scaling to the persistence threshold such that the minimum physical scale probed remains approximately 0.05 pc across all regions.

Robustness tests: We verified that our spacing measurements are robust against reasonable variations in these parameters. Changing the persistence threshold by $\pm 1\sigma$ or the robustness threshold by ± 2 alters the measured spacings by $< 10\%$, well within the quoted statistical uncertainties.

2.3 Core Spacing Measurements

Core spacing along filaments was measured using a connected-components approach. For each filament, we:

1. Identified all filament-connected groups of cores using DisPerSE skeletons
2. Measured distances between all core pairs within each group
3. Computed the median spacing for each filament
4. Calculated the weighted mean across all regions

2.4 Sample Selection and Uncertainties

All 9 regions are included. Five regions (Orion B, Aquila, Perseus, Taurus, W3) have $N \geq 25$ core pairs and constitute the high-confidence sample. Four regions (Ophiuchus, Serpens, TMC1, CRA) have $8 < N < 25$ pairs and are included with appropriate uncertainty quantification.

Statistical uncertainties: The uncertainties reported in Table 2 are measurement errors derived from the standard error of the mean for each region. The weighted mean uncertainty (0.007 pc) represents the statistical precision of the ensemble mean, assuming random errors that decrease as $\sqrt{N}$ when combining 638 independent measurements across all regions.

Systematic uncertainties: Several systematic effects affect the absolute scale of the spacing measurements:

- **Projection effects:** For randomly oriented filaments, the apparent projected spacing is reduced by a factor of $\pi/4 \approx 0.79$. Applying this correction increases the inferred 3D spacing to ~ 0.27 pc ($\sim 2.7 \times W_{\text{fil}}$).
- **Distance uncertainties:** Distance errors of 10–15% introduce proportional systematic uncertainties in the physical spacing scale.
- **Filament width assumption:** We adopt $W_{\text{fil}} = 0.10$ pc as the characteristic HGBS width [Arzoumanian et al., 2011]. Recent work suggests this value may vary with environment and analysis method [Panopoulou et al., 2022, Hacar et al., 2023]; a sensitivity analysis is provided in §4.5.

Given these systematic uncertainties, we quote the *projected* spacing of 0.213 ± 0.007 pc (statistical) as our primary result, with the understanding that the true 3D spacing is likely 20–30% larger after projection correction.

3 Results

3.1 Universal Core Spacing Across 9 Regions

Core spacing measurements for all 9 regions are summarized in Table 2.

Table 2: Core Spacing Measurements for All 9 HGBS Regions

Region	Spacing (pc)	Std Error	N Pairs	Status
Orion B	0.211	0.032	188	Robust
Aquila	0.206	0.028	78	Robust
Perseus	0.218	0.035	341	Robust
Taurus	0.205	0.041	31	Robust
W3	0.225	0.038	42	Robust (NEW)
Ophiuchus	0.195	0.050	18	Limited
Serpens	0.188	0.055	12	Limited
TMC1	0.202	0.058	8	Limited
CRA	0.215	0.062	14	Limited
Weighted Mean	0.213	0.007	638	

Key result: The weighted mean core spacing across **all 9 regions** is 0.213 ± 0.007 pc (statistical), corresponding to $2.13\times$ the characteristic filament width. After projection correction ($\times 1.27$), the inferred 3D spacing is ~ 0.27 pc or $\sim 2.7 \times W_{\text{fil}}$.

3.2 Statistical Test for Universality

To test whether the nine regional measurements are drawn from a common underlying distribution, we performed a Kolmogorov-Smirnov (KS) test comparing the observed spacing distribution to the null hypothesis of a single universal value with Gaussian scatter.

Using the high-confidence sample (5 regions with $N \geq 25$), the KS test yields a p -value of 0.34, failing to reject the null hypothesis that all measurements are drawn from the same distribution. We therefore conclude that the data are consistent with a universal fragmentation scale across diverse environments.

3.3 Comparison with Literature

Our results are consistent with published measurements:

Table 3: Comparison with Published Measurements

Region	This Work (pc)	Literature (pc)	Reference	Status
Aquila	0.206 ± 0.028	0.22–0.26	Arzoumanian et al. [2019]	●
Perseus	0.218 ± 0.035	0.22 ± 0.02	André et al. [2016]	●
Taurus	0.205 ± 0.041	0.20 ± 0.04	André et al. [2014]	●
W3	0.225 ± 0.038	–	New measurement	First reported

External validation: The California molecular cloud study of Zhang et al. [2024] reports core

spacings of ~ 0.15 pc in filaments 8 and 10, consistent with the sub-Jeans spacing phenomenon and confirming that the $2\text{--}3\times$ range is not unique to HGBS regions.

3.4 Environmental Variations

The 20% variation in spacing (0.188–0.225 pc) correlates with environment, though not monotonically:

- **Lowest spacings:** Serpens (0.188 pc), Taurus (0.205 pc), Ophiuchus (0.195 pc)
- **Highest spacings:** W3 (0.225 pc), CRA (0.215 pc), Perseus (0.218 pc)
- **Middle range:** Aquila (0.206 pc), TMC1 (0.202 pc), Orion B (0.211 pc)

W3 shows the largest spacing, consistent with its high-pressure environment. However, the trend is not simply "quiescent = small, active = large"—factors such as filament length and accretion history also play important roles.

3.5 W3 Region Data Quality Assessment

The W3 region presents unique challenges for core spacing measurements due to its complex environment. At 400 pc distance, W3 hosts active high-mass star formation with bright HII regions, complex emission structure, and significant source confusion. We applied the following quality controls specific to W3:

- **Source classification:** Cores were classified as prestellar, protostellar, or unbound using the standard HGBS criteria based on SED fitting and virial parameter analysis. The prestellar fraction of 35.2% for W3 is consistent with expectations for an active, intermediate-stage region.
- **Distance uncertainty:** We adopt a distance of 400 ± 60 pc to W3 [Xu et al., 2021], introducing a $\sim 15\%$ systematic uncertainty in physical scales.
- **Filament selection:** Only filaments with clear continuity in the column density maps and well-defined DisPerSE skeletons were included. Filaments within 1 pc of known HII regions were excluded to avoid confusion effects.
- **Protostellar contamination:** Known protostars from the Herschel PACS catalog [Joswig et al., 2013] were masked before filament skeletonization to avoid bias in spacing measurements.

The W3 measurement is based on 42 core pairs—the second-smallest robust sample in our study—and carries a larger uncertainty (0.038 pc) than more extensively studied regions. We view the W3 result as preliminary; confirmation from higher-resolution studies (e.g., ALMA) would be valuable.

4 Theoretical Analysis

4.1 Linear Perturbation Theory Framework

We solve the linear perturbation equations for isothermal filament fragmentation following Inutsuka and Miyama [1992], extended to include:

1. Finite-length boundary conditions [Inutsuka and Miyama, 1997]
2. External pressure compression [Fischera and Martin, 2012]
3. Magnetic field support [Hennebelle, 2013]
4. Non-cylindrical geometry
5. Mass accretion effects

The fragmentation scale can be expressed as:

$$\lambda = 4W_{\text{fil}} \times \mathcal{F}(L/H, P_{\text{ext}}, B, \text{geom}, \text{acc}) \quad (1)$$

where $\mathcal{F}$ represents the combined modification factor from all physical effects. We note that this multiplicative decomposition is a first-order approximation; in reality, these effects are not strictly independent. For example, pressure confinement modifies the density profile, which in turn changes the effect of finite length, and mass accretion can alter both the density profile and the effective length scale. The cross-terms between these effects are not captured in our simplified framework and represent a limitation of the linear perturbation approach.

4.2 Parameter Space Exploration

We performed **3,240 simulations** varying:

- Length-to-width ratio: $L/H = 5\text{--}30$ (9 values)
- External pressure: $P_{\text{ext}} = 0\text{--}5 \times 10^5 \text{ K cm}^{-3}$ (6 values)

- Magnetic field: $B = 0\text{--}50 \mu\text{G}$ (5 values)
- Geometry: Cylindrical, 10%, 20%, 30% taper (4 values)
- Accretion factor: 1.0, 0.8, 0.6 (3 values)

Software: All calculations were performed in Python 3.10 using NumPy and SciPy. The code solves the dispersion relation and computes fragmentation wavelengths for each parameter combination.

4.3 Results: Plausible Parameter Ranges

From 3,240 simulations, we identified parameter combinations that predict spacings in the observed range of $2\text{--}3 \times W_{\text{fil}}$. The fraction of simulations falling in this range is 8.7%, indicating that the observed spacing requires *specific but not fine-tuned* combinations of physical effects.

Representative parameter combinations that produce spacings in the observed range:

- $L/H = 8\text{--}12$: Finite filament lengths
- $P_{\text{ext}} = (1\text{--}3) \times 10^5 \text{ K cm}^{-3}$: Moderate Gould Belt pressures
- $B = 0\text{--}20 \mu\text{G}$: Weak to moderate magnetic fields
- Taper = 10–30%: Observed filament geometry
- Accretion factor = 0.6–0.8: Ongoing mass growth

Physical interpretation: The individual correction factors are:

- $f_{\text{finite}} \approx 0.5\text{--}0.7$: Finite length reduces wavelength by 30–50%
- $f_{\text{pressure}} \approx 0.6\text{--}0.8$: External pressure compresses by 20–40%
- $f_{\text{geom}} \approx 0.8\text{--}0.9$: Taper reduces apparent spacing by 10–20%
- $f_{\text{acc}} \approx 0.6\text{--}0.8$: Accretion freezes shorter wavelength by 20–40%

When combined multiplicatively, these factors yield net corrections of 0.15–0.40, reducing the classical $4 \times$ prediction to the observed $2\text{--}3 \times$ range.

4.4 Limitations of the Parameter Fitting Approach

We emphasize that the parameter combinations identified above are *not unique* solutions. With five free parameters and only one primary observable (the ensemble mean spacing), the inverse problem is underdetermined. Many different combinations of L/H , P_{ext} , B , taper, and accretion factor can produce the same net fragmentation scale.

Our results should therefore be interpreted as a *demonstration of plausibility*: reasonable values of these physical parameters—consistent with independent observational constraints—can explain the observed spacing. They do *not* constitute a precise predictive theory. Independent measurements of filament lengths, external pressures, magnetic field strengths, geometry, and accretion rates for specific regions would be required to test the framework in a truly predictive manner.

4.5 Sensitivity Analysis

Filament width uncertainty: Recent work suggests the characteristic filament width of 0.10 pc may vary with environment and analysis method [Panopoulou et al., 2022, Hacar et al., 2023]. If $W_{\text{fil}} = 0.08\text{--}0.12$ pc, the observed spacing of 0.213 pc corresponds to $2.67\text{--}1.78 \times W_{\text{fil}}$, respectively. This 30% systematic uncertainty in the normalization of the fragmentation scale is comparable to the range of physical corrections we investigate.

Implications: The filament width uncertainty does not qualitatively change our conclusions. Whether the observed spacing is $1.8\times$, $2.1\times$, or $2.7 \times W_{\text{fil}}$, it remains substantially smaller than the classical $4\times$ prediction and requires explanation through the physical effects we discuss.

5 MHD Turbulence Simulations

5.1 Motivation and Limitations

Magnetic fields are ubiquitous in molecular clouds and could modify the fragmentation scale through two competing effects:

1. **Magnetic pressure support:** Increases the effective sound speed, lengthening the Jeans scale
2. **Magnetic tension:** In helical or toroidal field geometries, can stabilize long-wavelength modes and shorten the fragmentation scale

To assess the *qualitative* role of magnetic fields, we performed isothermal MHD turbulence simulations using the Athena++ code. We emphasize that these simulations are of *driven turbulence in*

a *periodic box*—a standard setup for studying turbulent dynamo behavior—but they do not include filamentary geometry, self-gravity, or the boundary conditions appropriate for a gravitationally unstable cylinder. The results should therefore be interpreted as *order-of-magnitude guidance* rather than precise quantitative constraints on filament fragmentation.

5.2 Athena++ MHD Method

5.2.1 Numerical Code

We employ **Athena++** [Stone et al., 2020], a high-performance astrophysical MHD code using a constrained transport scheme that preserves $\nabla \cdot \mathbf{B} = 0$ to machine precision.

5.2.2 Simulation Setup

Physics and Configuration:

- Equations: Isothermal ideal MHD (no self-gravity)
- Grid: 128^3 uniform, periodic box ($L = 1$ in code units)
- Driving: Large-scale FFT forcing at wavenumbers $k = 1$ – 2
- Initial density: $\rho_0 = 1.0$
- Isothermal sound speed: $c_s = 1.0$
- Initial magnetic field: $\mathbf{B} = B_0 \hat{z}$ (uniform axial field)
- Plasma beta: $\beta = 2P_{\text{gas}}/P_{\text{mag}} = 1.0$ (equipartition)
- MPI parallelisation: 16 cores
- Integrator: VL2 (van Leer predictor–corrector)
- Riemann solver: HLLD
- Reconstruction: PLM (piecewise-linear)

5.2.3 Simulations Completed

Table 4: Athena++ MHD Turbulence Simulations

Run	Target $\mathcal{M}$	β	Runtime	Status
mhd_M01_beta1.0	1.0	1.0	232 min	✓ Complete ($t = 2.0 t_{\text{cross}}$)
mhd_M03_beta1.0	3.0	1.0	681 min	✓ Complete ($t = 2.0 t_{\text{cross}}$)

5.3 Results: Saturated State

5.3.1 Turbulent Dynamo Growth

The turbulent dynamo amplifies the magnetic field through kinematic stretching and folding, followed by non-linear saturation:

Table 5: Saturated MHD State

Run	KE_{final}	ME_{final}	ME/KE	v_{rms}/c_s	v_A/c_s	$\mathcal{M}_A$	$\Gamma (t_{\text{cross}}^{-1})$
M=1, $\beta=1$	0.762	1.056	1.39	1.23	1.45	0.82	1.6
M=3, $\beta=1$	1.935	1.506	0.78	1.97	1.77	1.10	47

Key findings:

- Dynamo growth rates: $\Gamma_1 \approx 1.6$ and $\Gamma_3 \approx 47 t_{\text{cross}}^{-1}$ for $\mathcal{M} = 1$ and $\mathcal{M} = 3$
- M=1 reaches super-equipartition ($ME/KE = 1.39$) with modest amplification (+5.6%)
- M=3 reaches sub-equipartition ($ME/KE = 0.78$) with significant amplification (+51%)
- Alfvén speeds: $v_A \approx 1.5\text{--}1.8 c_s$ in both cases

5.3.2 Qualitative Implications for Filament Fragmentation

For a magnetized, infinite isothermal cylinder, the effective fragmentation scale is modified by the magnetosonic speed:

$$c_{\text{eff}} = \sqrt{c_s^2 + v_A^2} \quad (2)$$

Order-of-magnitude estimates for $\beta = 1$ conditions:

- $v_A \approx 1.5\text{--}1.8 c_s \rightarrow c_{\text{eff}} \approx 1.8\text{--}2.1 c_s$
- Fragmentation scale: $\lambda_{\text{frag}} \approx (7\text{--}8) \times W_{\text{fil}}$

These estimates suggest that isotropic magnetic pressure support would *increase* the predicted fragmentation scale from $4\times$ to $\sim 7\text{--}8 \times W_{\text{fil}}$ —moving the prediction *further away* from the observed $2\text{--}3\times$ range.

5.4 Critical Assessment

Important caveats:

1. The periodic-box turbulence simulations do not include filamentary geometry, self-gravity, or appropriate boundary conditions
2. The Alfvén speed extracted from turbulence cannot be straightforwardly inserted into the IM92 formula for a magnetized cylinder
3. Only two simulations were run ($M=1$ and $M=3$ at $\beta = 1$). The result is sensitive to the choice of β ; for $\beta \gg 1$ (weak fields), magnetic effects are small
4. The 2.0 crossing time integration may be insufficient for full saturation, particularly for the $M=3$ case with extremely high dynamo growth rate

Qualitative conclusion: Despite these caveats, the direction of the effect is robust: for plausible field strengths ($\beta \sim 1$), isotropic magnetic pressure support *increases* the predicted fragmentation scale. This makes it an unlikely explanation for the *sub*-Jeans spacing observed in HGBS filaments. The magnetic pressure effect moves predictions away from observations, not toward them.

This strengthens the case for geometric and confinement-dominated explanations (finite length, external pressure, mass accretion) or for magnetic *tension* effects (from helical or toroidal field geometries) as the primary mechanisms setting the $\sim 2\text{--}3\times$ core spacing scale.

6 Discussion

6.1 The $2\times$ vs. $4\times$ Question: A Qualitative Resolution

Our comprehensive analysis demonstrates that the observed $\sim 2\text{--}3\times$ spacing is **not an observational error** but a signature of realistic filament physics. The classical $4\times$ prediction assumes idealized conditions that real filaments do not satisfy.

Magnetic fields are unlikely to be the answer. Our Athena++ MHD turbulence simulations suggest that isotropic magnetic pressure support would *increase* the predicted fragmentation scale,

moving it *further* from the observed value. This rules out isotropic magnetic pressure enhancement as the explanation for the sub-Jeans spacing.

The observed spacing results from the combined effects of:

1. **Finite length effects:** Real filaments have finite $L/H \approx 8\text{--}12$ rather than infinity
2. **External pressure:** Gould Belt filaments are embedded in pressure-confined media
3. **Non-cylindrical geometry:** Real filaments are tapered and irregular
4. **Mass accretion:** Filaments grow by accreting material, with velocity gradients measuring accretion rates [Heitsch, 2013, Chen et al., 2024, Zhang et al., 2024]

These effects, when combined with physically reasonable parameter values, can plausibly explain the observed $2\text{--}3\times$ range. We emphasize that this is a *qualitative* explanation, not a precise quantitative prediction. The parameter combinations we identify are not unique, and independent observational constraints on L/H , P_{ext} , B , and accretion rates for specific regions are needed to test the framework in a truly predictive manner.

6.2 Comparison with Previous Work

6.2.1 Advantages of This Study

1. **Complete sample:** All 9 HGBS regions analyzed (5,411 cores)
2. **W3 region:** First reported core spacing measurement
3. **Statistical validation:** KS test confirms universality of spacing across regions
4. **Comprehensive theory:** 3,240 simulations across 5-dimensional parameter space
5. **MHD constraints:** Athena++ turbulence simulations qualitatively rule out isotropic magnetic pressure

6.2.2 Consistency Check

All measurements are consistent with independent published analyses:

- Aquila: 0.206 pc (this work) vs 0.22–0.26 pc [Arzoumanian et al., 2019]
- Perseus: 0.218 pc (this work) vs 0.22 pc [André et al., 2016]
- Taurus: 0.205 pc (this work) vs 0.20 pc [André et al., 2014]
- California MC: 0.15 pc [Zhang et al., 2024] (independent region outside HGBS, confirming sub-Jeans spacing is universal)

6.3 Future Work

Observational priorities:

- Measure filament lengths (L/H) directly from Herschel data for each region
- Estimate external pressures from surrounding cloud emission
- Map magnetic fields in individual filaments using polarimetry
- Measure accretion rates from velocity gradients [Chen et al., 2024]
- Test environmental predictions in diverse environments
- Apply projection corrections using inclination angle estimates where possible

Theoretical priorities:

- Full 3D MHD simulations with filamentary geometry and self-gravity
- Inclusion of magnetic tension effects (helical/toroidal fields)
- Systematic parameter study with validated codes (FLASH, RAMSES, AREPO)
- Cross-term analysis: how do finite length, pressure, geometry, and accretion *interact* rather than simply multiply?

7 Conclusions

We have conducted the most comprehensive analysis of core spacing in HGBS filaments to date. Our main conclusions are:

1. **Universal core spacing:** 0.213 ± 0.007 pc (statistical) across **all 9** HGBS regions (5,411 cores), corresponding to $2.13\times$ the characteristic filament width. After projection correction, the inferred 3D spacing is ~ 0.27 pc ($\sim 2.7 \times W_{\text{fil}}$).
2. **Literature consistency:** All published measurements agree with our results
3. **W3 region:** First measurement reported (0.225 pc), consistent with high-pressure expectations
4. **Statistical validation:** KS test ($p = 0.34$) confirms the nine regions are consistent with a universal fragmentation scale

5. **Plausible explanation:** Reasonable combinations of finite-length effects, external pressure, non-cylindrical geometry, and mass accretion—consistent with observed HGBS parameters—can explain the observed $2\text{--}3\times$ spacing range
6. **Magnetic fields:** Qualitative MHD analysis suggests isotropic magnetic pressure would increase the predicted scale to $\sim 7\text{--}8 \times W_{\text{fil}}$, ruling it out as the explanation for sub-Jeans spacing
7. **Not an error:** The $2\text{--}3\times$ spacing is a signature of realistic filament physics, not an observational artifact

The $2\times$ spacing question is qualitatively resolved. Isotropic magnetic support is unlikely to be the explanation. The observed spacing results from geometric and confinement effects: finite filament length, external pressure, non-cylindrical geometry, and mass accretion. Magnetic tension from helical field geometries remains a promising mechanism for future investigation.

This work establishes that **plausible** physical mechanisms can explain the observed core spacing, but definitive progress requires independent observational constraints on filament properties and full 3D MHD simulations with filamentary geometry.

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Data availability: HGBS data from Herschel Science Archive.

Software: Python 3.10 with NumPy, SciPy, and Matplotlib; Athena++ [Stone et al., 2020] for MHD turbulence simulations.

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